

Yulong Zhao

Statistics PhD Candidate | Statistical Machine Learning & Sequential Modeling

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Summary

Statistics PhD candidate specializing in statistical machine learning for sequential and behavioral data. Built predictive and latent-variable models that substantially improved accuracy, log loss, and early-failure detection on large-scale real-world datasets.

Technical Skills

Programming: Python, R, SQL.

ML / Statistics: sequence modeling (GRU, TCN, Transformer), latent-variable models, EM, regularized GLMs, kernel methods, time-series forecasting.

Libraries / Tools: NumPy, pandas, scikit-learn, Numba, TensorFlow/Keras, Git, Slurm/HPC, LaTeX.

Model Evaluation: cross-validation, bootstrap, threshold calibration, multiple testing, simulation studies.

Selected Research & Machine Learning Projects

University of Arizona — PhD Researcher 2023 – Present

Predictive Modeling of Sequential User Behavior Sept 2023 – Present

- Built an **interpretable predictive model** for timestamped user-event sequences that captures both which actions users take and when they take them, including time-varying behavioral effects missed by count-based features.
- Improved 10-fold cross-validated accuracy from 93.5% to **96.1%** and **cut log loss by 50%** ($0.237 \rightarrow 0.118$) on 19,089 events from 2,668 users, outperforming the count-feature baseline on every fold.
- Identified **timing-sensitive behavioral signals** whose predictive effect reverses across the task, revealing information fundamentally lost by conventional aggregate features.

Unsupervised Skill and State Discovery from User Sequences 2026 – Present

- Designed a **structured latent-variable model** that jointly learns stable user skills and evolving behavioral states from multistep interaction sequences.
- Developed **sparse skill-to-behavior mappings** that learn which latent skills drive actions, interface choices, and state transitions, instead of requiring those relationships to be specified a priori.
- Applied the framework to 1,340 users in a multi-interface problem-solving task, **recovering interpretable** skills, behavioral states, and skill-action relationships without manual state labeling.

Real-Time Early Warning with a Controlled False-Alarm Rate 2026 – Present

- Built an **online early-warning framework** that continuously updates failure risk from partial user-action sequences, enabling intervention before task completion; developed concurrently with a 2026 **JASA** paper on the same core idea.
- Designed and **calibrated decision thresholds** to maximize early detection while explicitly controlling the false-alarm rate at $\leq 5\%$.
- Benchmarked GRU, temporal CNN, Transformer, and leakage-safe logistic baselines; the best temporal CNN detected **98.9%** of failing users at precision ≥ 0.98 while keeping false alarms below 4% across all three tasks, compared with 85.6% detection for GRU.

Multivariate Time-Series Forecasting & Regime-Shift Detection May 2024 – Present

- Built a **leakage-free walk-forward forecasting** pipeline using PCA and RBF-SVR, **reducing MSE by 38%** versus the historical-mean baseline and outperforming it in all 18 evaluation settings.
- Developed a multivariate **regime-shift detection** pipeline using recurrence-based features and multiple-testing correction, identifying statistically significant environmental shifts and separating robust signals from preprocessing-sensitive findings.

Education

University of Arizona — PhD, Statistics Expected May 2028

University of Wisconsin–Madison — MS, Statistics and Data Science 2023

University of Science and Technology of China — BS, Mathematics and Applied Mathematics 2022

Selected Publication & Presentation

Zhao, Y. & Tang, X. “Scalar-on-Process Model for Educational Assessment.” *Manuscript ready for submission.*

Refereed oral presentation, Symposium on Data Science & Statistics (SDSS), 2026.

Awards

Research Fellowship (2024–26), Incoming PhD Award, Academic Excellence Award, Outstanding Undergraduate Award.